Energy and Speed Are Related

Chapter 3 · Week 3

BIG QUESTION: How are energy and speed related?

VOCABULARY + DRAW

speed — Speed describes how _____ an object moves.

energy of motion — A moving object has energy of _____.

DRAW: the SAME ball moving slowly, then the SAME ball moving faster.

SLOWER (small arrow) **FASTER (long arrow)**

draw the same ball twice — use a longer arrow for faster

(Arrows are a model. Arrow length shows relative speed — not a measured speed.)

CORE STATEMENT

For the SAME OBJECT:

faster motion is associated with greater energy of motion.

Ramp Data + Graph

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NOTICE

EVIDENCE (our class data)

Record the average cup displacement for each ramp height.

Ramp Height	Average Cup Displacement (cm)
1 book	
2 books	
3 books	

GRAPH SUMMARY

Draw 3 bars (one per ramp height). Title: Ramp Height vs. Average Cup Displacement.

bars: 1 book · 2 books · 3 books (tallest bar at 3 books)

MEASURED vs INFERRED

WHAT DID WE MEASURE?

ramp height (books) · cup displacement (cm)

WHAT DID WE INFER?

relative speed · the speed↔energy relationship

We did NOT measure speed or energy — we measured the change and inferred the relationship.

Did We Measure Speed or Energy?

Chapter 3 · Week 3 · lift the flap

LIFT THE FLAP

DID WE DIRECTLY MEASURE SPEED OR ENERGY?

FOLD HERE (fold down, then lift to read)

NO.

We changed ramp height and measured the change caused to the target (cup displacement).
The pattern in our evidence helps us INFER the relationship between speed and energy of motion.

Remember the chain:

same object → faster motion → greater change

NO-CUT OPTION (flat fill-in)

Did we directly measure speed or energy? _____ (NO)
We measured the _____ (change) caused by the ball and _____ (inferred) the relationship.